

The Salyut programme itself ran for many more years. Salyut 2, Salyut 3 and Salyut 5, which quickly followed, would be part of the military-oriented Almaz programme.

Salyut 6 would be the first major advance in the space station programme. It and the Salyut 7, which closely resembled its predecessor, would be known as 'second-generation' Soviet space stations.

On December 10, 1977, Yuri Romanenko and Georgi Grechko became the first crew to board the space station. On March 12, 1981 the last crew – Vladimir Kovalyonok and Viktor Savinyikh – arrived and stayed for 75 days.

With Salyut 6, the Soviet space station program evolved from short-duration to long-duration stays. It was launched unmanned and crews arrived later in Soyuz spacecraft. It had two docking ports. This permitted refuelling and resupply by automated spacecrafts. They docked automatically at the aft port, and were then opened and unlocked by cosmonauts on the station. Transfer of fuel to the station took place automatically under supervision from the ground.

A second docking port also meant long-duration resident crews could receive visitors. Visiting crews often included cosmonaut-researchers from Soviet bloc countries or countries sympathetic to the Soviet Union. Vladimir Remek of Czechoslovakia, the first space traveler not from the US or USSR, visited Salyut 6 in 1978.

An experimental transport logistics spacecraft called Cosmos 1267 docked with Salyut 6 in 1982, proving that large modules could dock automatically with space stations – a major step toward the multimodular Mir station and the International Space Station.

The first crew of the Salyut 7 – Anatoli Berezovoi and Valentin Lebedev – arrived on May 13, 1982 aboard a Soyuz spacecraft.

The Salyut 7 carried three solar panels, two in lateral and one in dorsal longitudinal positions, but they now had the ability to mount secondary panels on their sides. Internally, the Salyut 7 carried electric stoves, a refrigerator, constant hot water and redesigned seats at the command console (more like bicycle seats). Two portholes were designed to allow ultraviolet light in, to help kill infections. Further, the medical, biological and exercise sections were improved, to allow long stays in the station. The BST-1M telescope used in Salyut 6 was replaced by an X-ray detection system.

On May 6, 1986 Leonid Kizim and Vladimir Solovyov became the last cosmonauts to visit the Salyut 7 space station, after which the programme was scrapped.